Egg Segmentation and Video Tracking

Nosh Robotics — Computer Vision Intern Assignment

Prepared: 29 October 2026

Project links

Public GitHub repository: https://github.com/Pavani1506/nosh-robotics-cv

Video tracking output (Google Drive): Open egg_tracking_v2.mp4

1. Objective and approach

The assignment explores egg segmentation with a pretrained Segment Anything Model (SAM), classical image-processing baselines, SAM-guided GrabCut refinement, evaluation against selected SAM pseudo-reference masks, and object tracking on the assigned video.

2. Dataset and SAM processing

- Dataset inventory: 110 images across the supplied dataset folders.
- SAM completed successfully on all 110 images using the SAM ViT-B checkpoint on Apple MPS.
- Recorded SAM parameters: points_per_side=8, pred_iou_thresh=0.86, stability_score_thresh=0.90, crop_n_layers=0, min_mask_region_area=100.
- SAM outputs include proposal masks and metadata. These are generated proposals, not manually verified egg ground-truth masks.

3. Classical methods and experiments

Implemented and explored Otsu thresholding, HSV thresholding with morphology, fixed-rectangle GrabCut, SAM proposal filtering, and SAM-guided GrabCut refinement. Batch scripts save masks, overlays, and CSV summaries. Experiment parameters and observations are recorded in reports/experiment_log.csv. Visual review showed both useful egg masks and failure cases, including background/package objects, missed eggs, and images where no candidate was selected.

4. Split and evaluation results

Item	Current result
Provisional split	88 development / 22 evaluation images (approx. 4:1)
Holdout predictions	22 of 22 evaluation images processed
Matching SAM pseudo-references	14 images compared; 8 missing matching references
Mean IoU	0.9573 on the 14 matched comparisons
Mean Dice	0.9767 on the 14 matched comparisons

Important interpretation: IoU and Dice above measure agreement with automatically selected SAM pseudo-reference masks, not verified ground-truth accuracy. Eight evaluation images had no matching selected SAM reference in the current comparison. The split was created after earlier experimentation, and the filename grouping did not prove absence of near-duplicate leakage; therefore it should be treated as provisional rather than a fully independent benchmark.

5. Video tracking

A baseline OpenCV tracker uses HSV/color thresholding, morphology, contour filtering, and centroid-based association. The V2 output is available through the Google Drive link above. Visual review showed that eggs were captured in sampled frames, but occasional false detections (including fingernails/reflections) remain. Tracking is therefore reported as a working baseline, not a fully validated tracker.

6. Repository and reproducibility

- The public GitHub repository contains source scripts, README, requirements.txt, experiment log, and split manifest.
- The dataset, generated outputs, and SAM checkpoint are not included in the repository. Follow the README to obtain the dataset/checkpoint and set up the environment.
- AI tools assisted with implementation and debugging; code and limitations should be reviewed and understood.

7. Remaining limitations / next improvements

- Create or obtain manually verified egg ground-truth masks for a defensible accuracy evaluation.
- Generate matching SAM pseudo-reference selections for all evaluation images and report coverage explicitly.
- Repeat evaluation with a split established before tuning, with duplicate/near-duplicate review where feasible.
- Improve candidate selection and video tracking to reduce false positives and missed eggs.

This report describes the current state of the project and its known limitations. Results are preliminary and are not presented as verified ground-truth performance.